Rohan Nargundkar

Embedded Systems | Firmware Development

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Github,Website

Summary

Electronics and Communication Engineering student with an interest in embedded systems, firmware development, device drivers and digital electronics. Hands-on experience developing C-based peripheral simulations and working with microcontrollers, sensor interfaces and embedded monitoring applications

Projects

Smart Industrial IoT System for Energy and Vibration Monitoring | Ongoing

ESP32, ADXL345, INA219, I2C, Embedded C, MicroSD

- Designing an embedded condition-monitoring system that tracks machine vibration and electrical parameters to enable early fault detection and predictive maintenance
- Interfacing an ADXL345 accelerometer and INA219 power monitor with an ESP32 over I2C, using Embedded C firmware
- Building real-time data acquisition and MicroSD logging of vibration and power consumption for trend analysis

Virtual GPIO Driver and Hardware Simulator | Completed

Embedded C, Device Drivers, Unit Testing

- Developed a software-based GPIO peripheral model and driver in Embedded C, simulating hardware registers without physical hardware
- Implemented pin configuration, digital input/output, and interrupt handling, with structured error handling for invalid operations
- Built an automated test suite validating the driver, achieving 12/12 test cases passed

Skills

Programming Languages: C, Embedded C, Python

Embedded Systems: Microcontroller Programming, Sensor Interfacing, GPIO, Interrupt Handling, Serial Communication, I2C, UART, PWM Fundamentals

Digital Electronics: Combinational and Sequential Logic Design, Digital Circuit Simulation, Waveform Analysis

Analog Electronics: Analog and Digital Circuit Design, Circuit Testing and Debugging

Software and Tools: Arduino IDE, Keil µVision, Cadence, GTKWave, Make, CMake, Git, VS Code

Platforms: ESP32, ATmega328P, Linux

Education

Bachelor of Engineering – Electronics and Communication Engineering

Jain College of Engineering, Belagavi **2023-2027**

CGPA-7.92

Pre-University Course – PCMB
Govindram Seksaria Science (GSS) College, Belagavi **2021–2023**
Percentage: **83%**

SSLC – State Board
MVM's Prabhatai Deshpande English Medium School, Belagavi **2009–2021**
Percentage: **89%**

Training & Workshops

Cadence EDA Workshop

- Gained practical exposure to industry-standard Cadence EDA tools, including Virtuoso, Innovus and Genus.
- Explored Cadence Virtuoso for schematic design, circuit simulation and custom IC layout concepts.
- Learned the fundamentals of Cadence Genus for RTL synthesis and understanding the conversion of RTL descriptions into gate-level netlists.
- Explored Cadence Innovus for digital physical design concepts, including floorplanning, placement, clock tree synthesis and routing.
- Developed an understanding of the overall VLSI design flow, from RTL design and synthesis to physical implementation.

Areas of Interest

Embedded Firmware Development | Embedded C Programming | Microcontroller Architecture | VLSI Design | Industrial IoT | Digital Electronics